

Scalability & Future Plan

Date	15 July 2026
Team ID	
Project Name	Book A Doctor
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited language support	Restricts accessibility for non-English speaking patients	High
2	Notification service slows under heavy load	Delays in SMS/email reminders during peak usage	Medium
3	No telemedicine module	Patients cannot consult doctors remotely	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports ~1,000 concurrent users	Scale backend with AWS Auto-Scaling, implement load balancer
2	Data Storage	MongoDB Atlas with single cluster	Upgrade to sharded clusters for horizontal scaling
3	Performance	Average response time ~2 seconds	Introduce caching (Redis), optimize queries, use CDN for static assets
4	Security	Basic JWT authentication	Add multi-factor authentication, role-based access control, and regular penetration testing

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-language support	Aug 2026	Wider accessibility for diverse patient groups
Phase 3	Telemedicine video consultation	Sep 2026	Enables remote healthcare, reduces hospital crowding
Phase 4	AI-powered doctor recommendation	Oct 2026	Personalized patient experience, faster doctor selection
Phase 5	Integration with wearable health devices	Nov 2026	Preventive healthcare insights and real-time monitoring